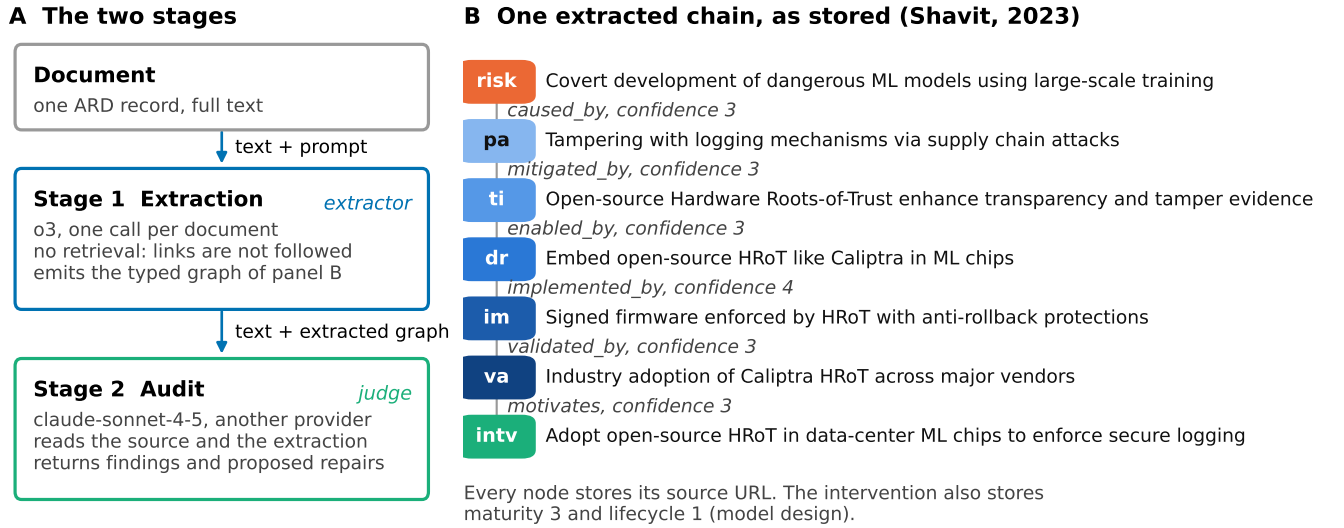
A The two stages
Document
one ARD record, full text
text + prompt
Stage 1 Extraction
extractor
o3, one call per document
no retrieval: links are not followed
emits the typed graph of panel B
text + extracted graph
Stage 2 Audit
judge
claude-sonnet-4-5, another provider
reads the source and the extraction
returns findings and proposed repairs
B One extracted chain, as stored (Shavit, 2023)
risk
Covert development of dangerous ML models using large-scale training
caused_by, confidence 3
pa
Tampering with logging mechanisms via supply chain attacks
mitigated_by, confidence 3
ti
Open-source Hardware Roots-of-Trust enhance transparency and tamper evidence
enabled_by, confidence 3
dr
Embed open-source HRoT like Caliptra in ML chips
implemented_by, confidence 4
im
Signed firmware enforced by HRoT with anti-rollback protections
validated_by, confidence 3
va
Industry adoption of Caliptra HRoT across major vendors
motivates, confidence 3
intv
Adopt open-source HRoT in data-center ML chips to enforce secure logging
Every node stores its source URL. The intervention also stores
maturity 3 and lifecycle 1 (model design).